

Student identification

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Registration number	C11/26/FCDF/17155
Linux distribution / version	Kali Linux
VM hostname	Kali

Lab 21: Using the Shell

```
(bevamanfu@kali)-[~]
$ echo $SHELL
/usr/bin/zsh

(bevamanfu@kali)-[~]
$ pwd
/home/bevamanfu

(bevamanfu@kali)-[~]
$ ls
archive  Desktop  Downloads  Pictures  Templates  wadf-challenge  week1_command_history.txt
backup   Documents Music      Public    Videos    wadf-labs

(bevamanfu@kali)-[~]
$ ls -l
total 52
drwxr-xr-x 2 bevamanfu kali 4096 Jul 17 22:18 archive
drwxr-xr-x 2 bevamanfu kali 4096 Jul 17 22:24 backup
drwxr-xr-x 2 bevamanfu kali 4096 Jul 18 00:22 Desktop
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Documents
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Downloads
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Music
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Pictures
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Public
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Templates
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Videos
drwxr-xr-x 3 bevamanfu kali 4096 Jul 17 23:04 wadf-challenge
drwxr-xr-x 6 bevamanfu kali 4096 Jul 17 17:21 wadf-labs
-rw-r--r-- 1 bevamanfu kali 632 Jun 26 08:19 week1_command_history.txt

(bevamanfu@kali)-[~]
$ cd Documents

(bevamanfu@kali)-[~/Documents]
$ pwd
/home/bevamanfu/Documents

(bevamanfu@kali)-[~/Documents]
$ cd ~

(bevamanfu@kali)-[~]
$ pwd
/home/bevamanfu
```

```

(bevamanfu@kali)~$ cd ..

(bevamanfu@kali)~/home$ pwd
/home

(bevamanfu@kali)~/home$ history 12
 1 hostnamectl
 2 uname -r
 3 sudo passwd root
 4 hostnamectl
 5 uname -r
 6 cat /etc/os-release
 7 lsblk
 8 df -h
 9 whoami
10 echo "$SHELL"
11 date
12 timedatectl list-timezones

(bevamanfu@kali)~/home$ history |tail -n 12
105 pwd
106 echo $SHELL
107 pwd
108 ls
109 ls -l
110 cd Documents
111 pwd
112 cd ~
113 pwd
114 cd ..
115 pwd
116 history 12

```

Lab 22: Configuring the Shell

```

(bevamanfu@kali)~/home$ cat ~/.zshrc
# ~/.zshrc file for zsh interactive shells.
# see /usr/share/doc/zsh/examples/zshrc for examples

setopt autocd           # change directory just by typing its name
#setopt correct         # auto correct mistakes
setopt interactivecomments # allow comments in interactive mode
setopt magicequalsubst   # enable filename expansion for arguments of the form 'anything=expression'
setopt nonomatch         # hide error message if there is no match for the pattern
setopt notify            # report the status of background jobs immediately
setopt numericglob       # sort filenames numerically when it makes sense
setopt promptsubst       # enable command substitution in prompt

WORDCHARS='_-' # Don't consider certain characters part of the word

# hide EOL sign ('%')
PROMPT_EOL_MARK=""

# configure key bindings
bindkey -e                                # emacs key bindings
bindkey ' ' magic-space                   # do history expansion on space
bindkey '^U' backward-kill-line           # ctrl + U
bindkey '^[[3;5~' kill-word               # ctrl + Supr
bindkey '^[[3~' delete-char               # delete
bindkey '^[[1;5C' forward-word             # ctrl + →
bindkey '^[[1;5D' backward-word           # ctrl + ←
bindkey '^[[5~' beginning-of-buffer-or-history # page up
bindkey '^[[6~' end-of-buffer-or-history   # page down
bindkey '^[[H' beginning-of-line          # home
bindkey '^[[F' end-of-line                # end
bindkey '^[[Z' undo                       # shift + tab undo last action

# enable completion features
autoload -Uz compinit
compinit -d ~/.cache/zcompdump
zstyle ':completion:*:*:*:*' menu select
zstyle ':completion:*' auto-description 'specify: %d'

```

```

(bevamanfu@kali)-[/home]
$ env
COLORFGBG=15;0
COLORTERM=truecolor
COMMAND_NOT_FOUND_INSTALL_PROMPT=1
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
DESKTOP_SESSION=lightdm-xsession
DISPLAY=:0.0
DOTNET_CLI_TELEMETRY_OPTOUT=1
GDMSESSION=lightdm-xsession
HOME=/home/bevamanfu
LANG=en_US.UTF-8
LANGUAGE=
LOGNAME=bevamanfu
NMAP_PRIVILEGED=
PANEL_GDK_CORE_DEVICE_EVENTS=0
PATH=/home/bevamanfu/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr/games:/home/bevamanfu/.dotnet/tools
POWERSHELL_TELEMETRY_OPTOUT=1
POWERSHELL_UPDATECHECK=Off
PWD=/home
QT_ACCESSIBILITY=1
QT_AUTO_SCREEN_SCALE_FACTOR=0
QT_QPA_PLATFORMTHEME=qt5ct
SESSION_MANAGER=local/kali:@/tmp/.ICE-unix/1429,unix/kali:/tmp/.ICE-unix/1429
SHELL=/usr/bin/zsh
SHLVL=1
SSH_AGENT_PID=1529
SSH_AUTH_SOCK=/home/bevamanfu/.ssh/agent/s.RDJpj98jBr.agent.yibv7eigIb
TERM=xterm-256color
USER=bevamanfu
WINDOWID=0
XAUTHORITY=/home/bevamanfu/.Xauthority

```

```

(bevamanfu@kali)-[/home]
$ alias ll='ls -l'

(bevamanfu@kali)-[/home]
$ ll
total 84
drwx----- 5 Abena      is          4096 Jul 17 15:59 Abena
drwx----- 5 adwoa      labgroup    4096 Jul 17 13:22 adwoa
drwx----- 5 Akosua     Geography   4096 Jul 17 20:23 Akosua
drwx----- 5 Ama       engineering 4096 Jul 17 15:58 Ama
drwx----- 5 Nate       OffSec     4096 Jul 17 18:29 Araba
drwx----- 5 Ethan      OffSec     4096 Jul 17 18:34 Bev
drwx----- 20 bevamanfu kali        4096 Jul 23 22:36 bevamanfu
drwx----- 5 Akosua     group1     4096 Jul 17 19:17 Elinam
drwx----- 5 eng_admin   engineering 4096 Jul 17 15:53 eng_admin
drwx----- 5 Ethan      OffSec     4096 Jul 17 20:23 Ethan
drwx----- 5 is_admin    is         4096 Jul 17 15:54 is_admin
drwx----- 5 John       sales      4096 Jul 17 15:59 John
drwx----- 5 Kofi       is         4096 Jul 17 15:59 Kofi
drwx----- 5 Kojo       engineering 4096 Jul 17 15:58 Kojo
drwx----- 5 Ethan      OffSec     4096 Jul 17 20:11 Minette
drwx----- 5 Akosua     Geography   4096 Jul 17 20:00 Nate
drwx----- 5 Nate       IT         4096 Jul 17 20:07 Nathan
drwx----- 5 sales_admin sales      4096 Jul 17 15:54 sales_admin
drwx----- 5 Akosua     Geography   4096 Jul 17 20:13 Simon
drwx----- 5 user1      group1     4096 Jul 17 20:30 user1
drwx----- 5 Yaa       sales      4096 Jul 17 15:58 Yaa

(bevamanfu@kali)-[/home]
$ nano ~/.zshrc

```

```

Session Actions Edit View Help
GNU nano 8.7.1 /home/bevamanfu/.zshrc
alias egrep='egrep --color=auto'
alias diff='diff --color=auto'
alias ip='ip --color=auto'

export LESS_TERMCAP_mb='${\E[1;31m}' # begin blink
export LESS_TERMCAP_md='${\E[1;36m}' # begin bold
export LESS_TERMCAP_me='${\E[0m}' # reset bold/blink
export LESS_TERMCAP_so='${\E[01;33m}' # begin reverse video
export LESS_TERMCAP_se='${\E[0m}' # reset reverse video
export LESS_TERMCAP_us='${\E[1;32m}' # begin underline
export LESS_TERMCAP_ue='${\E[0m}' # reset underline
export MANROFFOPT="-c"

# Take advantage of $LS_COLORS for completion as well
zstyle ':completion:*' list-colors "${(s.:.)LS_COLORS}"
zstyle ':completion:*:kill:*:processes' list-colors '=(#b) #([0-9])#*=0=01;31'
fi

# some more ls aliases
alias ll='ls -l'
alias la='ls -A'
alias l='ls -CF'

# enable auto-suggestions based on the history
if [ -f /usr/share/zsh-autosuggestions/zsh-autosuggestions.zsh ]; then
    . /usr/share/zsh-autosuggestions/zsh-autosuggestions.zsh
    # change suggestion color
    ZSH_AUTOSUGGEST_HIGHLIGHT_STYLE='fg=244'
fi

# enable command-not-found if installed
if [ -f /etc/zsh_command_not_found ]; then
    . /etc/zsh_command_not_found
fi

alias ll='ls -l'

```

```

(bevamanfu@kali)-[/home]
$ source ~/.zshrc

(bevamanfu@kali)-[/home]
$ alias
diff='diff --color=auto'
egrep='egrep --color=auto'
fgrep='fgrep --color=auto'
grep='grep --color=auto'
history='history 0'
ip='ip --color=auto'
l='ls -CF'
la='ls -A'
ll='ls -l'
ls='ls --color=auto'
which-command=whence

(bevamanfu@kali)-[/home]
$ ll
total 84
drwx----- 5 Abena is 4096 Jul 17 15:59 Abena
drwx----- 5 adwoa labgroup 4096 Jul 17 13:22 adwoa
drwx----- 5 Akosua Geography 4096 Jul 17 20:23 Akosua
drwx----- 5 Ama engineering 4096 Jul 17 15:58 Ama
drwx----- 5 Nate OffSec 4096 Jul 17 18:29 Araba
drwx----- 5 Ethan OffSec 4096 Jul 17 18:34 Bev
drwx----- 20 bevamanfu kali 4096 Jul 23 22:59 bevamanfu
drwx----- 5 Akosua group1 4096 Jul 17 19:17 Elinam
drwx----- 5 eng_admin engineering 4096 Jul 17 15:53 eng_admin
drwx----- 5 Ethan OffSec 4096 Jul 17 20:23 Ethan
drwx----- 5 is_admin is 4096 Jul 17 15:54 is_admin
drwx----- 5 John sales 4096 Jul 17 15:59 John
drwx----- 5 Kofi is 4096 Jul 17 15:59 Kofi
drwx----- 5 Kojo engineering 4096 Jul 17 15:58 Kojo
drwx----- 5 Ethan OffSec 4096 Jul 17 20:11 Minette
drwx----- 5 Akosua Geography 4096 Jul 17 20:00 Nate
drwx----- 5 Nate IT 4096 Jul 17 20:07 Nathan
drwx----- 5 sales_admin sales 4096 Jul 17 15:54 sales_admin
drwx----- 5 Akosua Geography 4096 Jul 17 20:13 Simon
drwx----- 5 user1 group1 4096 Jul 17 20:30 user1

```

```

(bevamanfu@kali)-[/home]
$ cd ~

(bevamanfu@kali)-[~]
$ ll
total 52
drwxr-xr-x 2 bevamanfu kali 4096 Jul 17 22:18 archive
drwxr-xr-x 2 bevamanfu kali 4096 Jul 17 22:24 backup
drwxr-xr-x 2 bevamanfu kali 4096 Jul 18 00:22 Desktop
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Documents
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Downloads
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Music
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Pictures
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Public
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Templates
drwxr-xr-x 2 bevamanfu kali 4096 Jun 26 07:28 Videos
drwxr-xr-x 3 bevamanfu kali 4096 Jul 17 23:04 wadf-challenge
drwxr-xr-x 6 bevamanfu kali 4096 Jul 17 17:21 wadf-labs
-rw-r--r-- 1 bevamanfu kali 632 Jun 26 08:19 week1_command_history.txt

```

Lab 23: File Globbing

```

(bevamanfu@kali)-[/home]
$ cd ~

(bevamanfu@kali)-[~]
$ mkdir globbing_test

(bevamanfu@kali)-[~]
$ cd globbing_test

(bevamanfu@kali)-[~/globbing_test]
$ touch file1.txt file2.txt file3.txt report.txt data.csv notes.md

(bevamanfu@kali)-[~/globbing_test]
$ ls
data.csv file1.txt file2.txt file3.txt notes.md report.txt

(bevamanfu@kali)-[~/globbing_test]
$ ls *.txt
file1.txt file2.txt file3.txt report.txt

(bevamanfu@kali)-[~/globbing_test]
$ ls file*
file1.txt file2.txt file3.txt

(bevamanfu@kali)-[~/globbing_test]
$ ls file?.txt
file1.txt file2.txt file3.txt

(bevamanfu@kali)-[~/globbing_test]
$ ls file[12].txt
file1.txt file2.txt

```

```
(bevamanfu@kali)~/globbing_test
$ ls file[!3].txt

(bevamanfu@kali)~/globbing_test
$ ls file[sudo passwd root].txt
zsh: bad pattern: file[sudo

(bevamanfu@kali)~/globbing_test
$ ls file['!3'].txt
file3.txt

(bevamanfu@kali)~/globbing_test
$ ls file[\\!3].txt
file1.txt file2.txt

(bevamanfu@kali)~/globbing_test
$ ls file[^3].txt
file1.txt file2.txt
```

```
(bevamanfu@kali)~/globbing_test
$ ls *.{csv,txt}
data.csv file1.txt file2.txt file3.txt report.txt

(bevamanfu@kali)~/globbing_test
$ mkdir backup

(bevamanfu@kali)~/globbing_test
$ cp *.txt backup/

(bevamanfu@kali)~/globbing_test
$ ls backup/
file1.txt file2.txt file3.txt report.txt

(bevamanfu@kali)~/globbing_test
$
```

Lab 24: File Manipulation

```
(bevamanfu@kali)~/globbing_test
$ touch newfile1.txt newfile2.txt

(bevamanfu@kali)~/globbing_test
$ ls -l newfile*
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:25 newfile1.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:25 newfile2.txt

(bevamanfu@kali)~/globbing_test
$ echo "This is content"> myfile.txt

(bevamanfu@kali)~/globbing_test
$ cat myfile.txt
This is content

(bevamanfu@kali)~/globbing_test
$ echo "Additional Content" >> myfile.txt

(bevamanfu@kali)~/globbing_test
$ cat myfile.txt
This is content
Additional Content

(bevamanfu@kali)~/globbing_test
$ cp newfile1.txt newfile1_backup.txt

(bevamanfu@kali)~/globbing_test
$ ls -l newfile1*
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:27 newfile1_backup.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:25 newfile1.txt

(bevamanfu@kali)~/globbing_test
$ cp *.txt archive/
cp: target 'archive/': No such file or directory

(bevamanfu@kali)~/globbing_test
$ mkdir archive

(bevamanfu@kali)~/globbing_test
$ cp *.txt archive/
```

```
(bevamanfu@kali)~/globbing_test
$ ls archive/
file1.txt file2.txt file3.txt myfile.txt newfile1_backup.txt newfile1.txt newfile2.txt report.txt

(bevamanfu@kali)~/globbing_test
$ mv newfile1_backup.txt newfile1_backup_old.txt

(bevamanfu@kali)~/globbing_test
$ rm file2.txt

(bevamanfu@kali)~/globbing_test
$ ls file2.txt
ls: cannot access 'file2.txt': No such file or directory

(bevamanfu@kali)~/globbing_test
$ rm -i file3.txt
rm: remove regular empty file 'file3.txt'? y

(bevamanfu@kali)~/globbing_test
$ ll
total 12
drwxr-xr-x 2 bevamanfu kali 4096 Jul 24 02:29 archive
drwxr-xr-x 2 bevamanfu kali 4096 Jul 24 02:22 backup
-rw-r--r-- 1 bevamanfu kali 0 Jul 23 23:48 data.csv
-rw-r--r-- 1 bevamanfu kali 0 Jul 23 23:48 file1.txt
-rw-r--r-- 1 bevamanfu kali 35 Jul 24 02:26 myfile.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:27 newfile1_backup_old.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:25 newfile1.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:25 newfile2.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 23 23:48 notes.md
-rw-r--r-- 1 bevamanfu kali 0 Jul 23 23:48 report.txt
```

```
(bevamanfu@kali)~/globbing_test]
$ ls -l backup/
total 0
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:22 file1.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:22 file2.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:22 file3.txt
-rw-r--r-- 1 bevamanfu kali 0 Jul 24 02:22 report.txt

(bevamanfu@kali)~/globbing_test]
$ rm -r backup

(bevamanfu@kali)~/globbing_test]
$ ls -l backup/
ls: cannot access 'backup/': No such file or directory

(bevamanfu@kali)~/globbing_test]
$
```